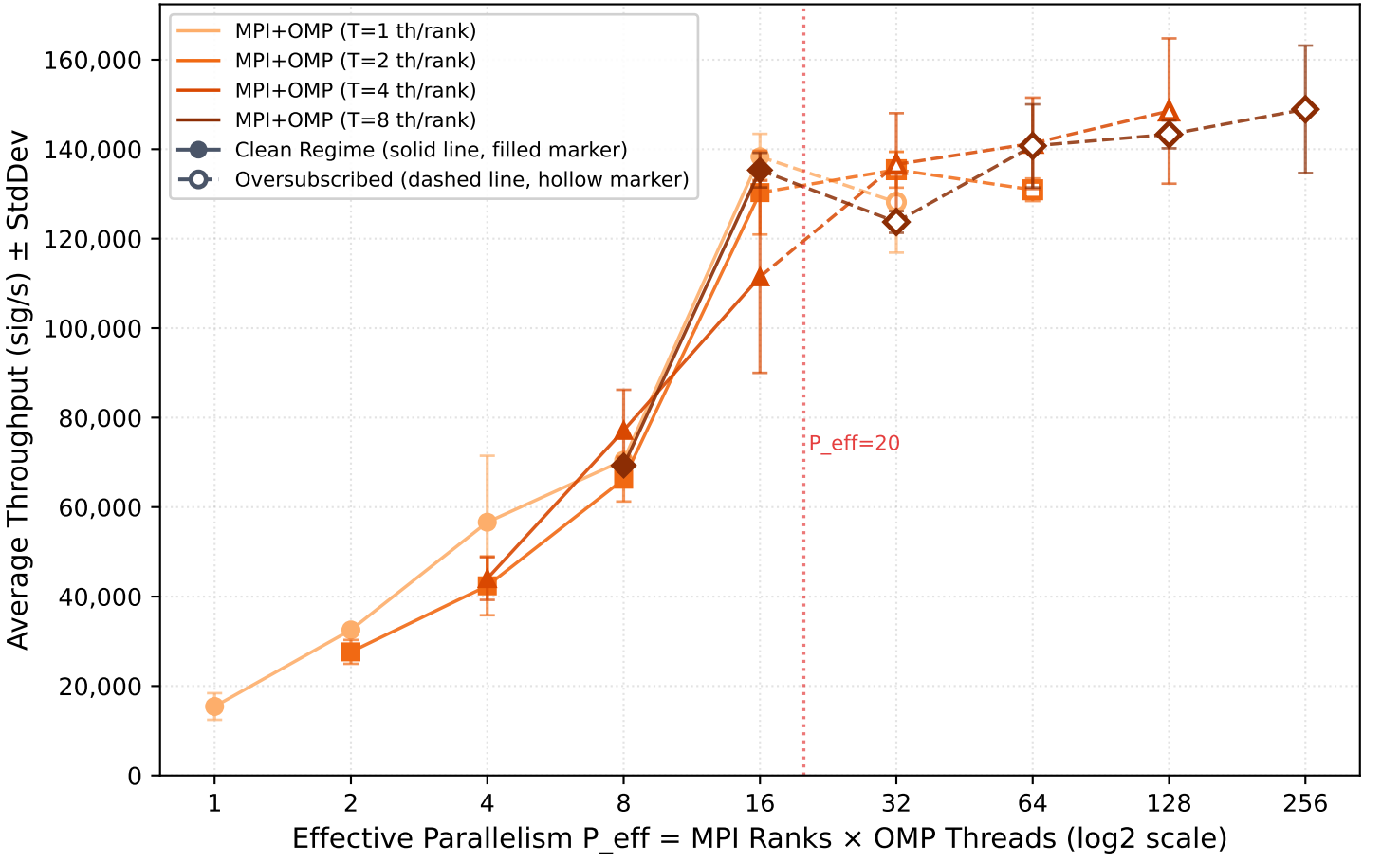
Task B: MPI+OpenMP Throughput vs P_eff with Error Bars (Batch = 5,000 signatures)
MPI+OMP (T=1 th/rank)
MPI+OMP (T=2 th/rank)
MPI+OMP (T=4 th/rank)
MPI+OMP (T=8 th/rank)
Clean Regime (solid line, filled marker)
Oversubscribed (dashed line, hollow marker)
P_eff=20
Average Throughput (sig/s) ± StdDev
0
20,000
40,000
60,000
80,000
100,000
120,000
140,000
160,000
1
2
4
8
16
32
64
128
256
Effective Parallelism P_eff = MPI Ranks × OMP Threads (log2 scale)